MAHATMA GANDHI UNIVERSITY
BCA (HONOURS) FOURTH SEMESTER PRACTICAL EXAMINATION
(2024 ADMISSION ONWARDS)
MG4SECBCA200 - Object Oriented Programming using Java

Instructions to Examiners

Duration of the Examination: 2 Hrs

Maximum Marks: 35

● Students must submit the Lab Record to attend the Semester End Examination.

● Lab question paper contains two questions one from the list of questions in Module 4 and the other from Module 5.

● Ask students to write the procedure/ Source code in Java for the assigned problem.

● Conduct a Viva-Voce(oral examination) for the course.

Evaluation Criteria:

Question 1 – Question from module 4 [A][CO4] (10 Marks)

Logic - 5 Marks, Successful Compilation - 2 Marks, Output - 3 Marks

Question 2 – Question from module 5 [A][CO5] (15 Marks)

Problem Understanding & Design **- 10 Marks,** Successful Compilation & Execution **- 5 Marks,**

Viva - 5 Marks

Record - 5 Marks

List of Questions

Question 1- Questions from module 4 [A] [CO4].

1. Write a Java program to read an integer from the user and print its factorial**.**
2. Write a Java program to read a number *n* and display the Fibonacci series up to *n* terms.
3. Write a Java program to find the sum of digits of a given number.
4. Write a Java program to count the number of even and odd digits in a number.
5. Write a Java program to read two numbers as command line arguments and print **their** sum, difference, product, and quotient**.**
6. Write a Java program to read a number from command line argument and check whether it is even or odd**.**
7. Write a Java program to read a number from command line argument and find its factorial**.**
8. Write a program to sort an integer array in ascending order using bubble sort
9. Write a program to perform linear search in an array.
10. Write a program to perform matrix multiplication
11. Write a program to find largest and smallest element in a matrix
12. Write a program to check whether the given string is a palindrome or not.
13. Programs to Utilize String Methods
14. Write a program to sort a string array in alphabetic order using bubble sort
15. Write a program to illustrate hierarchical inheritance
16. Write a program to illustrate method overriding
17. Write a program to illustrate dynamic method dispatch
18. Write a program to illustrate the use of an abstract class
19. Write a java application program to simple tasks using multithreading such as display even numbers and odd numbers, print multiplication table of 3 numbers etc. with suitable delay
20. Write a program to find cube of a number using packages
21. Write a program to find factorial of a number using packages
22. Write a java program to read n numbers and race an exception called NegativeException when you input a negative number?
23. Create a class Student with attributes roll no, name, age and course. If age of student is not with in a specific range then generate user-defined exception AgeNotWithinRangeException".
24. Write a program to implement multiple inheritance
25. Create a class Rectangle with overloaded constructors and method to calculate area: Default constructor (length = 1, breadth = 1), Constructor with one parameter (square) and Constructor with two parameters (rectangle)

Question 2- Questions from module 5 [A] [CO5].

1. Create a simple JFrame to display the message *"Welcome to Java Swing"* using a JLabel.
2. Create a JFrame with a button. On clicking the button, open a JDialog showing a message.
3. Create a login window using JFrame with username and password fields
4. Create a form using JLabel, JTextField, and JButton to accept student name and display it on button click.
5. Develop a program using JTextArea to enter feedback and display character count.
6. Create a program using JCheckBox to select hobbies and display selected hobbies.
7. Create a program using JRadioButton to select gender.
8. Create a program using JList to display a list of programming languages.
9. Create a program using JComboBox to select a country from the list.

10. Create a program using JTable to display student mark details
11. Write a swing program to accept an integer in a textbox then reverse that number and display the result in the second textbox?
12. Write a swing program to accept a value in a textbox then find the area of a circle and display the result in the second textbox?
13. Design a form using FlowLayout to arrange buttons.
14. Create a window using BorderLayout and place components in all five regions.
15. Design a calculator interface using GridLayout.
16. Create a form using GridBagLayout for student registration.
17. Design multiple screens using CardLayout (e.g., Login → Home).
18. Create a program using Null Layout and manually set component positions
19. Write a swing program to accept a value in a textbox then find the Volume of a sphere and display the result in the second textbox? (hint :V = 4/3 πr3)
20. Write a swing program to accept an integer in a textbox then find the sum of digits of that number and display the result in the second textbox?
21. Write a java program to accept a number then check whether a given number is positive or negative and display the result in the second textbox
22. Write a swing program to accept a value in a textbox then find the factorial of that number and display the result in the second textbox?
23. Write a program using swing to accept values in two textboxes then find the largest number and display the result in third text box.
24. Create a form using GridBagLayout for student registration.
25. Write a program to insert a record in database
26. Write a program to update a record in database
27. Write a program to delete a record in database